

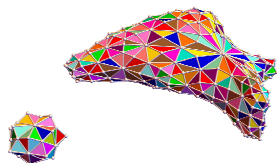
Index Formula

g : locally injective function on (V,E)
 $S_g(v) = \{g - g(v) = 0\} \subset S(v)$

If $i_g(v)$ is the Poincare-Hopf index,
 $j_g(v) = [i_g(v) + i_{-g}(v)]/2$
 is called symmetric index.

Theorem: (2012)

$$j_g(v) = 1 - \frac{\chi(S(v))}{2} - \frac{\chi(S_g(v))}{2}$$



We see random $S_g(v)$ in a 4-manifold.

Proof:

start with decomposition :

$$S(v) = S_g^+(v) \cup S_g^-(v) \cup S_g(v)$$



start with decomposition : $S_g(v)$ has dimension shifted, because edges are smallest units.

Euler characteristic:

$$\chi(S(v)) = \chi(S_g^+(v)) + \chi(S_g^-(v)) - \chi(S_g(v))$$

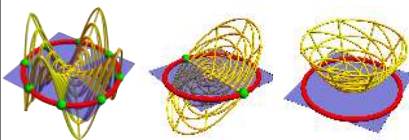
$$1 - \frac{\chi(S(v))}{2} = \frac{1}{2} - \frac{\chi(S_g^+(v))}{2} + \frac{1}{2} - \frac{\chi(S_g^-(v))}{2} + \frac{\chi(S_g(v))}{2}$$

$$1 - \frac{\chi(S(v))}{2} = j_g(v) + \frac{\chi(S_g(v))}{2}$$

Solve

$$j_g(v) = 1 - \frac{\chi(S(v))}{2} - \frac{\chi(S_g(v))}{2}$$

QED



$$j_g(v) = 1 - 0/2 - 8/2 = -3 \quad j_g(v) = 1 - 0/2 - 2/2 = 0 \quad j_g(v) = 1 - 0/2 - 0/2 = 1$$

Continuum

If $g : M \rightarrow \mathbb{R}$ is smooth function and v is a critical point, for a small enough sphere $S(v)$

$$j_g(v) = 1 - \frac{\chi(S(v))}{2} - \frac{\chi(S_g(v))}{2}$$

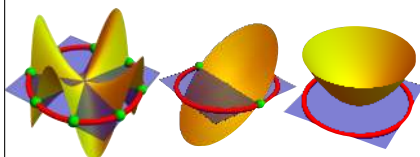
For odd dimensional manifolds,

$$j_g(v) = 1 - \frac{\chi(S(v))}{2} - \frac{\chi(S_g(v))}{2} = 0$$

For even dimensional manifolds,

$$j_g(v) = 1 - \frac{\chi(S(v))}{2} - \frac{\chi(S_g(v))}{2} = 1 - \frac{\chi(S_g(v))}{2}$$

Euler characteristic of a 4 manifold is the difference of the volume and a Hilbert-Einstein type action.



Odd dimensional manifolds are flat

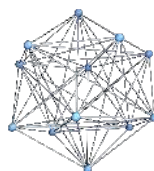
$k(v)$ index expectation curvature that is $g \leftrightarrow -g$ invariant

Odd dimensional manifolds with such curvature are flat.

Proof: $S_g(v)$ is odd-dim manifold

$$j_g(v) = 1 - \frac{\chi(S(v))}{2} - \frac{\chi(S_g(v))}{2}$$

are both zero for odd dim manifolds QED



For any 5-manifold:

$$K = -E/6 + F/4 - C/6$$

is constant 0.

$$f_G(t) = 1 + 12t + 60t^2 + 160t^3 + 240t^4 + 192t^5 + 64t^6$$

Curvature as an average genus

Define $1 - \chi(G)/2$ the genus of G .

For even dimensional manifolds: curvature is the average genus of the level set $S_g(v)$ in $S(v)$.

Proof: $S_g(v)$ is even-dim manifold

$$j_g(v) = 1 - \frac{\chi(S_g(v))}{2} - \frac{\chi(S(v))}{2}$$

genus of level set in $S(v)$ zero for even dim manifolds

curvature is an expectation of $j_g(v)$.



4 manifolds

$$\kappa(v) = E[1 - \frac{\chi(S_g(v))}{2}]$$

$$\chi(S_g(v)) = \sum_{w \in S_g(v)} K(w)$$

$\chi(G) - |V(G)|$ is expectation of curvatures of random 2D-surface curvatures. Compare: Hilbert-Einstein action is the expectation of sectional curvature of a random sheets.

